

# Principal Component Analysis

Unsupervised

## PCA

↳ PCA is a dimensionality reduction technique.

↳ It is a feature extraction algorithm, not feature selection.

↓  
Convert  $n$  feature into  $K$ , where  $K < n$

## Application of PCA

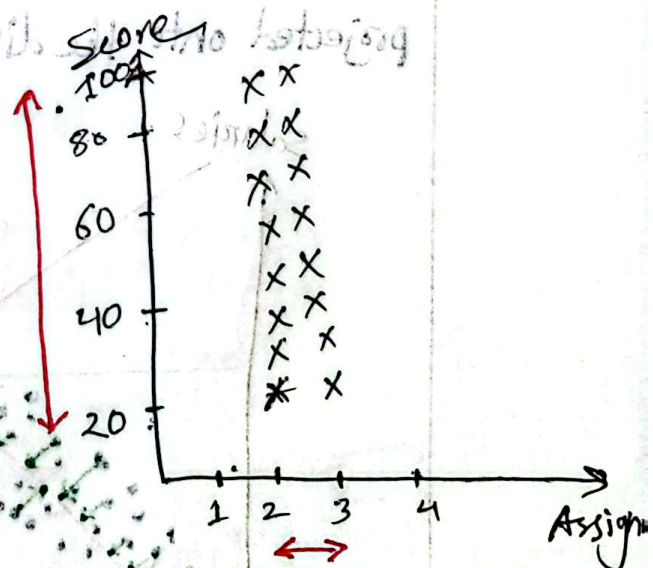
1. Data Compression → ~~Convert~~ Converting a 50D data into 10D
2. Data Visualization → reduce data to 2D or 3D, in order to visualize the dataset.
3. Speed Up Computation → reduce load on memory.

## Intuition of PCA

- Preserve the direction/feature having maximum variability.
- more spread = more information

Ex: 2D → 1D

| Score | Assignment | CGPA |
|-------|------------|------|
| 91    | 3          |      |
| 52    | 2          |      |
| 85    | 3          |      |
| 79    | 3          |      |
| 63    | 2          |      |
| 25    | 2          |      |

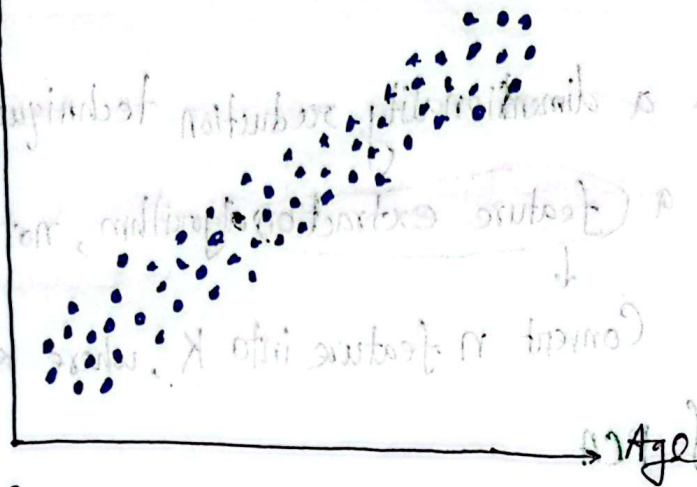


Since the spread of Assignment is less, so we can remove Assignment.



Ex-2

salaries



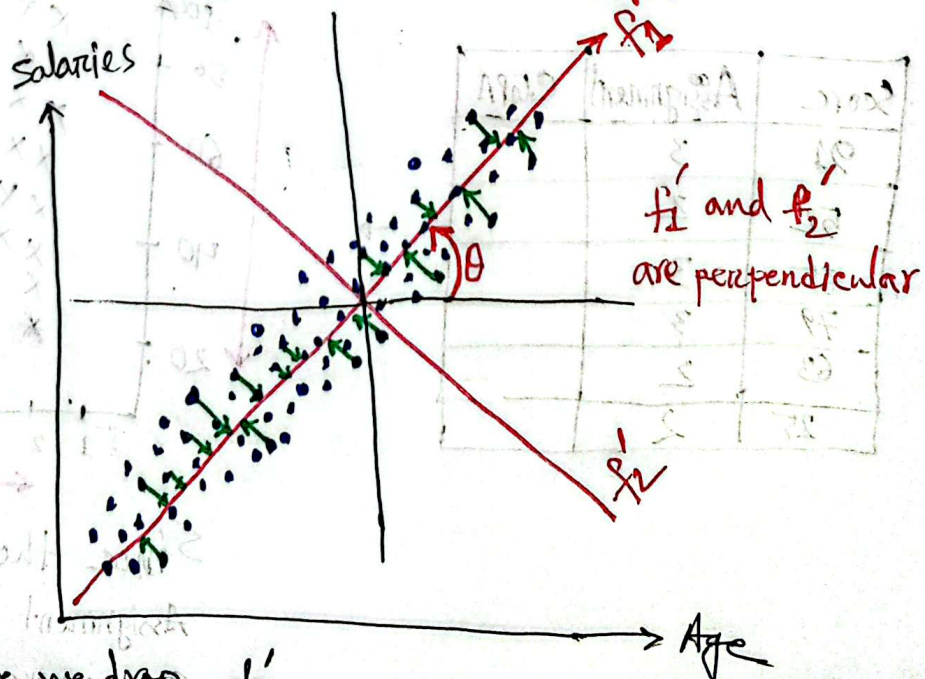
In this figure, both direction have almost same spread features. So,

- (i) Find the direction  $f'_1$  and  $f'_2$  such that  $f'_1 \gg f'_2$ .
- (ii) drop  $f'_2$ .
- (iii) Project all the points onto  $f'_1$ .

Thus  $2D \rightarrow 1D$

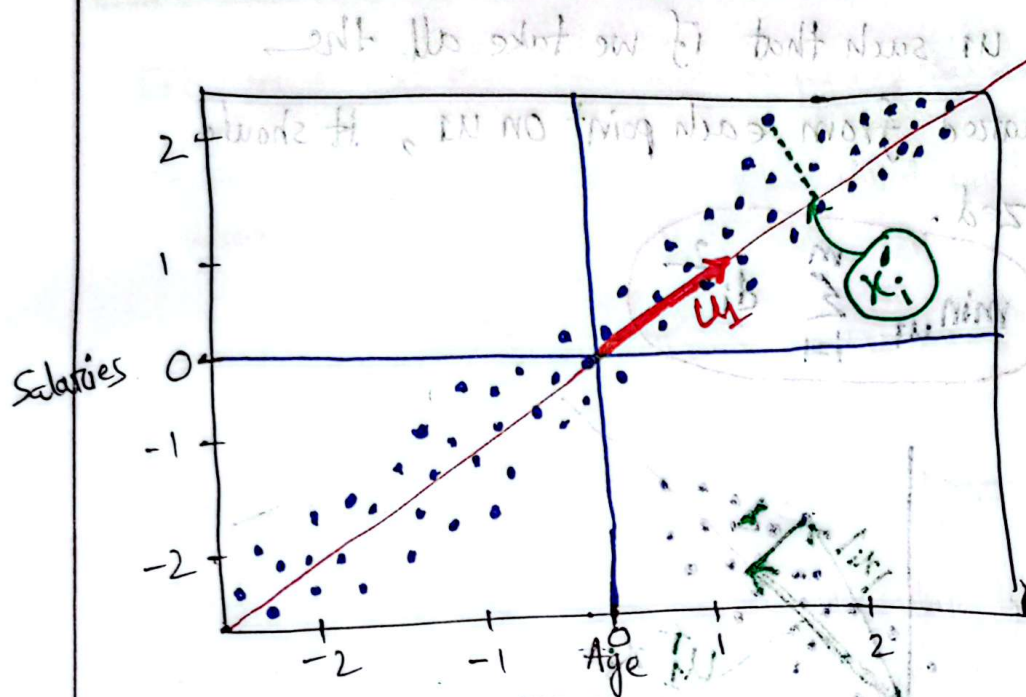
It is like the rotation of axis.

Note: Find a direction such that the variance of  $x$  projected onto the direction is maximum.



Here we drop,  $f'_2$

## PCA Objective - Maximizing Variance



Here,  $U_1$  = unit vector.

$$\text{So, } |U_1| = 1$$

$$\therefore x_i' = \text{proj}_{U_1}(x_i)$$

$$\Rightarrow x_i' = \frac{U_1 \cdot x_i}{|U_1|}$$

$$\Rightarrow x_i' = U_1 \cdot x_i$$

$$\Rightarrow x_i' = U_1^T \cdot x_i$$

$$\begin{matrix} 1 \times n & n \times 1 \\ \searrow & \swarrow \\ & 1 \times 1 \end{matrix}$$

⊗ Find  $U_1$  such that  $\text{var}\{\text{proj}_{U_1}(x_i)\}$  is maximum.

Now,  $D = \{x_i\}_{i=1}^m$  and  $D' = \{x_i'\}_{i=1}^m$

$\uparrow$  2D                       $\uparrow$  1D

Similarly,  $\bar{x}' = U_1^T (\bar{x})$

$\uparrow$  1D                       $\uparrow$  2D

$$\text{So, } \text{var}\{\text{proj}_{U_1}(x_i)\} = \frac{1}{m} \sum_{i=1}^m (U_1^T x_i - U_1^T \bar{x})^2$$

If mean of datasets = 0,  $= \frac{1}{m} \sum_{i=1}^m (U_1^T x_i)^2$

So, find such  $U_1$  so that

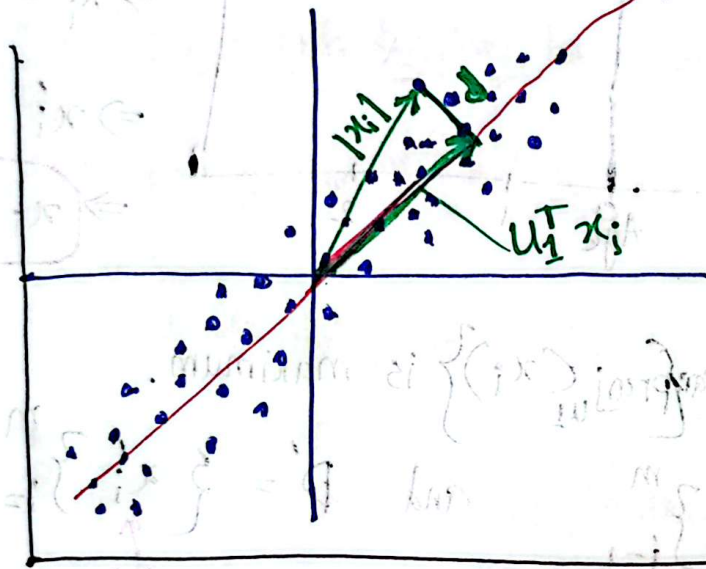
$$\frac{1}{m} \sum_{i=1}^m (U_1^T x_i)^2 \text{ is maximum.}$$



## PCA Objectives - Minimizing Distances

Goal  $\rightarrow$  Find  $u_1$  such that if we take all the distance squared from each point on  $u_1$ , it should be minimized.

$$\min_{u_1} \sum_{i=1}^m d_i^2$$



$$\therefore d_i^2 = |x_i|^2 - (u_1^T x_i)^2$$

$$\Rightarrow d_i^2 = x_i^T x_i - (u_1^T x_i)^2$$

$$\text{So, } \min_{u_1} \sum_{i=1}^m (x_i^T x_i - (u_1^T x_i)^2)$$

## Eigen Values & Vectors

Let input matrix,  $X = \begin{bmatrix} \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}_{m \times n}$

where  $X \rightarrow$  column standardization / mean centering,  $\mu = 0$

So, Covariance  $(X) = S \rightarrow \boxed{S = \frac{1}{n} \cdot X^T \cdot X}$   $(n, n)$

Defination: If  $S$  is an  $n \times n$  matrix, then a non-zero vector  $v$  in  $\mathbb{R}^n$  is called an eigen vector of  $S$  if  $SV$  is a scalar multiplication of  $V$ .

that is  ~~$SV = \lambda V$~~   $SV = \lambda V$  for some scalar  $\lambda$ .

Eigen value decomposition,

$$\text{Eigen values } (S) = \lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \dots \geq \lambda_n$$

$$\text{Eigen vectors } (S) = \begin{matrix} \downarrow & \downarrow & \downarrow & \dots & \downarrow \\ v_1 & v_2 & v_3 & \dots & v_n \end{matrix}$$

So,  $\lambda_1$  is maximum eigen values.

In our case,

$u_1 = v_1 =$  eigen vector of  $S (X^T X)$   
 $\nearrow$   
maximum variance direction



## PCA Summary

① Column Standardise

②  $S = \frac{X^T X}{n}$

③ Compute eigen values & vectors of  $S$

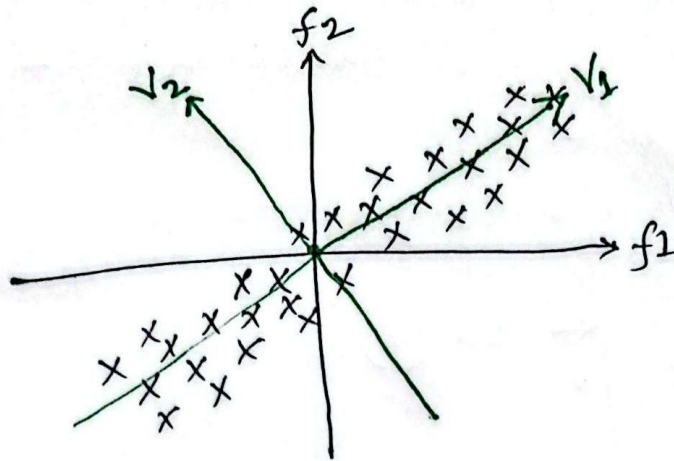
④  $u_1 = v_1, u_2 = v_2, \dots, u_n = v_n$

⑤ Select top  $K$  eigen vectors.

⑥ Project  $x_i$  on selected eigen vectors.

$$V^T X = V^T X$$

# Understanding Eigen-values - Geometrically



$X \rightarrow 2$  dimensions

So,  $n=2$

$\therefore S = (2, 2)$

$\text{eig}(S)$ ;

$\lambda_1 \geq \lambda_2$

$v_1 \perp v_2$

$\lambda \rightarrow$  gives us a relative scale of variance explaining.

When,  $X \in \mathbb{R}^{10}$ ,  $n=10$

$\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \dots \geq \lambda_{10}$

$\downarrow$   
 $v_1$

direction with  
max variance

$\downarrow$   
 $v_2$

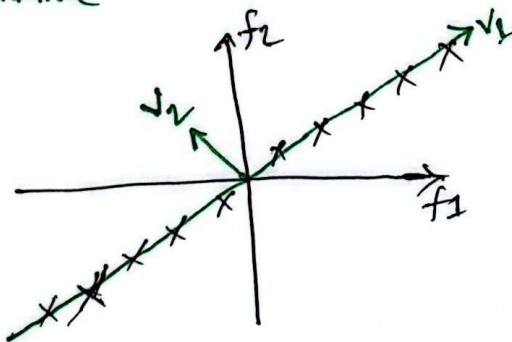
direction with  
2nd max variance

$\downarrow$   
 $v_3$

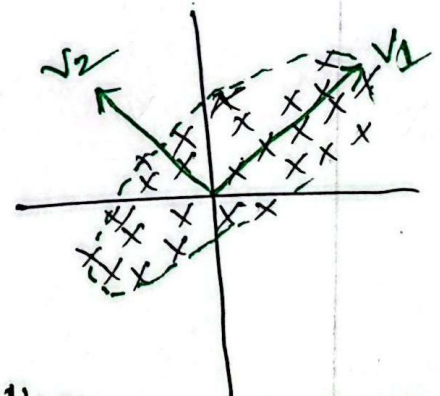
$\downarrow$   
 $v_{10}$

$10D \rightarrow 2D$   
 $X^{10} \rightarrow \tilde{X} \in \mathbb{R}^2$

Ex



Let,  $\lambda_1 = 5.8$  } Approx.  
 $\lambda_2 = 0$



Here,  $\lambda_1 = 4.9$  } Approx.  
 $\lambda_2 = 2.1$

Now, .  
Significance of  $\lambda_1 = \frac{\lambda_1}{\lambda_1 + \lambda_2} = 61\%$   
Significance of  $\lambda_2 = \frac{\lambda_2}{\lambda_1 + \lambda_2} = 39\%$